Permissions and Human Review

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach permissions with explicit policy cards and paper proposals.

Separate the worked policies, preserve the privacy extension and collect the inventory task before feedback.

Keep one habit: Ask which exact permission permits each action, and keep factual correctness separate.

MAIN LESSON · SECTION 1

Prepare the main lesson and its retained extension

> Use fictional records and schedule separate extension time.

Goal and preparation

Primary target: Explain the **mechanism** and terminology of **least privilege** and tool scopes

Two 45-minute sessions. **Least privilege** grants only capabilities required by the task. Tool access should include resource and operation scope, not merely a broad tool name.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

By the end, I can enforce **least privilege** by checking operation and object before execution.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Main plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval and readiness

Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of state machines for bounded agents

Reference example, not the only acceptable wording: Teacher explanation: A bounded agent uses explicit states and finite action limits. A state machine makes invalid transitions visible.

States are ready, looking_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised **evidence**. done is terminal. Maximum two lookup attempts.

Without opening the old lesson, explain one key idea from Prompt Experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of controlled prompt experiments

Reference example, not the only acceptable wording: Teacher explanation: A prompt experiment compares complete specifications using the same task cases.

Define success before inspecting **output**s; change one intended instruction and preserve other settings. Evaluate both prompts on paired cases with fixed model/version, source records, **output** contract and scoring rubric.

Use development results for selection, then reserve new final cases. Score each case rather than counting attractive sentences.

Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

Look for: Encode and decode text under an explicit toy vocabulary

Reference example, not the only acceptable wording: Teacher explanation: The rule chooses the longest matching ordinary token; sat has length three and s length one.

Vocabulary and matching are case sensitive and contain no uppercase C.

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

▼ Materials and extension planning

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

Apply the connected idea: Privacy and least privilege

Teach Week 29 as a two-session investigation using fictional data. A fictional class activity dashboard needs category counts, not names or personal messages. Design its data and role **permission**s before collection.

This exercise uses invented data and policy cards; do not change the Academy's actual authentication settings. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

No; names are not required for that calculation.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Case exposure check: this source example appears in ai-2/31, ai-6/6. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Agents and Tools

Prompt Experiments

Text as Pieces

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A teacher approves posting version 1 to a class board. The assistant changes the text and destination to a public board. Does the old approval cover the new action? Define what must be reviewed.

Concept to explain

Week 31: **Permission**s and Human Review

Term	Meaning
permission	Authorisation to perform a specified action on specified information or resources.
approval gate	A point where a specified action must wait for explicit authorisation.
consequence	An outcome or effect of a decision, including who benefits or is harmed.
least privilege	Giving a user or system only the permissions needed for its authorised task.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model operation and resource checks before execution

> Keep public-room, catalogue and role-policy examples separately labelled.

Define least privilege and the reporting-assistant policy

Least privilege grants only capabilities required by the task. Tool access should include resource and operation scope, not merely a broad tool name.

A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes. Check actor, operation and resource before executing a proposed call.

Trace the first scope violation

Policy dimension	Reporting assistant
Actor	The stated reporting assistant with its supplied trusted scope
Operation	lookup only
Resource	public-room records only
Other operations or resources	not permitted under this policy

Tool token permits lookup(public-room). A proposed call writes teacher notes.

Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

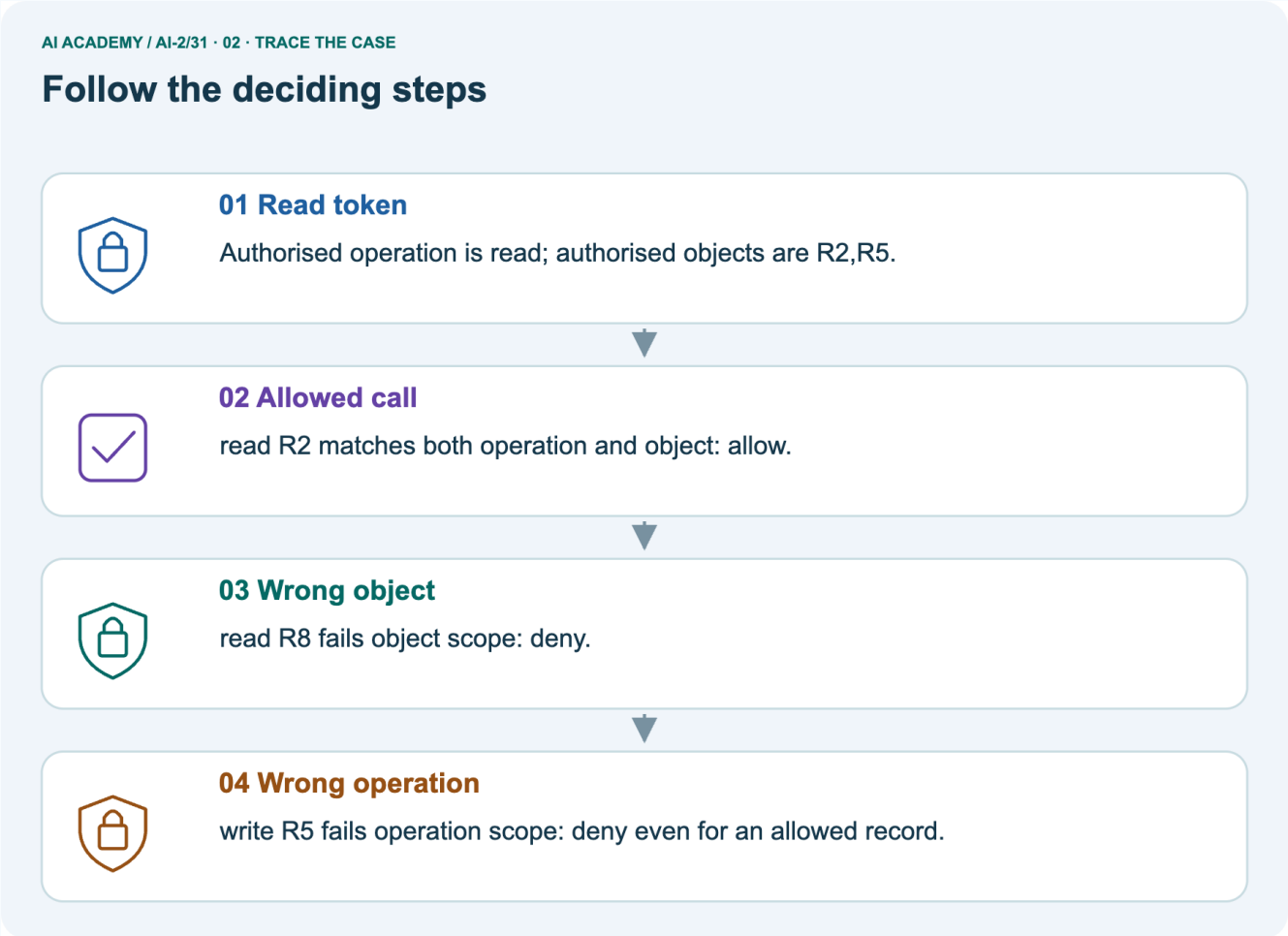
Read the separate catalogue token

A fictional catalogue helper receives a narrowly scoped tool token.

Token allows read on catalogue records R2 and R5 only. No write **permission** is granted.

Proposals: read R2; read R8; write R5. Generated text claims all are helpful.

Trace its proposals



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read token	Authorised operation is read; authorised objects are R2,R5.
2. Allowed call	read R2 matches both operation and object: allow.
3. Wrong object	read R8 fails object scope: deny.
4. Wrong operation	write R5 fails operation scope: deny even for an allowed record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Permissions are concrete restrictions, not a confidence score or a general instruction to be useful. Each proposed call must match both dimensions before it happens.

A request written inside catalogue text cannot enlarge the token's authority. Denial should preserve the proposed action and reason without executing the disallowed mutation.

Predict the requested scope change

Change: The helper says it needs write R5 to correct an apparent typo.

Prediction: The write remains denied under the existing token.

Reason: A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

Explain the permission-versus-correctness limit

Permission checks do not establish that a read result is trustworthy or that an allowed action is factually correct.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Privacy extension · Complete role code and server boundary

A fictional class activity dashboard needs category counts, not names or personal messages. Design its data and role **permission**s before collection.

This exercise uses invented data and policy cards; do not change the Academy's actual authentication settings.

```
roles = {"student": ["read_summary"],
         "teacher": ["read_summary", "read_review"]}
def allowed(role, action):
    return action in roles.get(role, [])

print(allowed("student", "read_summary"))
print(allowed("student", "read_review"))
print(allowed("unknown", "read_summary"))
```

The toy code prints True, False, False. It models a **permission** decision, not a complete security system.

A real server must verify identity and enforce **authorization** on every protected request; hiding a button in a page is insufficient.

No. The server must reject an unauthorized request even if a link is guessed. Exit: No. It omits authentication, server integration, session protection and many other controls; it demonstrates only a policy lookup.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Low **consequence**: suggest three destinations.

2. Higher **consequence**: contact families, spend money, or submit forms.

3. The system drafts; an authorized adult reviews and decides.

4. Log what was approved, rejected, or changed, then stop after the approved scope.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Posting a recommendation requires teacher review.

Place the **approval gate**.

Exact explanation to model

After draft and **evidence** are visible, before posting; include content and destination. No approval means no post.

Explain the added worked example

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Least privilege	Granting only the necessary authority
Capability	A scoped permission to perform an operation
Resource scope	The specific records an operation may address

Apply the connected idea: Privacy and least privilege

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Permissions and Human Review

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the main explanation, then fade support

> Collect each learner's reasoning before revealing a key.

Main completion question and key

Completion task: Tool token permits lookup(public-room). A proposed call writes teacher notes. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Main individual check

Use the worked case for **Least privilege** and tool scopes. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A generated plan cannot expand the token or actor scope.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed case

Which proposal is allowed, and why is an authorised object alone insufficient?

Only read R2. R5 is an allowed object for reading, but its write operation exceeds scope.

Changed-case reasoning: The write remains denied under the existing token. A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep learners explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Extension teaching cycle

Target: Collect only necessary information

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Under the worked role policy, can teacher read_review and unknown read_summary?

Teacher check: Teacher can; unknown cannot.

Individual check: The toy role-policy function returns True for a permitted read. Does that establish that every collected field is necessary and can be retained indefinitely? Explain what else must be decided.

Expected correction: No. A permission-table result does not justify unnecessary fields, authenticate the caller or specify **retention**. Choose fields for the stated purpose, use an approved **retention**/deletion process and name the responsible owner.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: Posting a recommendation requires teacher review.

Write the question to explain: Place the **approval gate**.

Reveal after **prediction**s: After draft and **evidence** are visible, before posting; include content and destination. No approval means no post.

Use the earlier transfer question as optional practice. Reserve Decide inventory calls independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: Both operation and resource exceed scope; deny before execution, regardless of generated confidence. Qualification: A generated plan cannot expand the token or actor scope.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

Stage	Teacher action
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Individual reasoning check

Connected teaching cycle: Privacy and least privilege

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied main investigation

> Keep the selected independent task out of shared practice.

Supplied records and answer guidance

Catalogue proposals: read R2 is allowed; read R8 fails object scope; write R5 and delete R2 fail operation scope. Reading R5 is allowed, but a write instruction in its contents does not authorise a write. Adult review must cover the exact draft, destination and cost before an action. All actions here are paper proposals.

Main W2 violates both operation and resource scope. Privacy code prints True, False, False; teacher/read_review is allowed and unknown/read_summary is denied. Keep the inventory task independent.

Predict, investigate and record

1. Predict → Test → Record → Explain

2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.

3. Audit the five supplied catalogue proposals under read R2/R5 only, then draw a fictional draft→adult review→permitted-action gate. Check exact content and destination. All decisions are paper simulations; do not execute messages, edits or purchases.

4. What did the hardest test teach you that the normal test did not?

Keep the evidence

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Main misconceptions

The worked result establishes every future case: A generated plan cannot expand the token or actor scope.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Choose support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Extension misconceptions

A successful run proves the conclusion: **Retention** must be tied to a purpose and school policy rather than a guessed universal number of days.

Identify who can correct or delete records and avoid keeping raw responses indefinitely just because storage is available.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: a library category counter.

Require category counts without names, default-deny **permission**s and an owner for **retention**/deletion decisions.

A convenient shortcut preserves meaning: A form asks for names, birthdays and comments to count activity preferences. Remove fields not needed for the purpose.

Aggregated counts can still expose information in very small groups; choose reporting rules with the responsible teacher.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied main investigation for shared practice. Reserve Decide inventory calls independently for the individual first attempt. The named extension has separate tasks and feedback.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Check these misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected main independent task

> Reveal its key after the first attempt; extension tasks have their own keys.

Selected independent question and key

Selected case: Scope allows read inventory items A, B. Calls read A, read C, write B. Decide.

Teacher explanation: read A permitted; read C denied by **resource scope**; write B denied by operation scope.

Judge reasoning and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Least privilege** grants only capabilities required by the task. Tool access should include resource and operation scope, not merely a broad tool name.

A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes. Check actor, operation and resource before executing a proposed call.

W2 Trace the worked case

Tool token permits lookup(public-room). A proposed call writes teacher notes. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

W3 Audit the limit

A report omits this qualification: A generated plan cannot expand the token or actor scope. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A generated plan cannot expand the token or actor scope. State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A generated plan cannot expand the token or actor scope. The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Extension questions and keys

Privacy and least privilege — W1 Minimize fields

For a count of preferred activities, identify which of activity choice, full name and birthday is necessary.

Activity choice is needed; full name and birthday are not needed for an anonymous category count.

Privacy and least privilege — W2 Trace permissions

Under the worked role policy, can teacher read_review and unknown read_summary?

Teacher can; unknown cannot.

Look up the requested action in the list for each role. The teacher list contains read_review, so that request is permitted. unknown has no policy entry; get supplies an empty list, which does not contain read_summary.

This models a **permission** decision after a role is trusted; it does not authenticate a visitor.

Privacy and least privilege — W3 Repair access control

A website hides teacher answers using CSS but sends them to student browsers. Explain the flaw and required boundary.

Students still receive the content. Enforce teacher **authorization** on the server and do not send answers in student responses.

Privacy and least privilege — W4 Design independently

Design a fictional preference summary with student and teacher roles. List necessary fields, permitted actions, a **retention** purpose and a deletion owner.

Use anonymous choice/count data, student access to approved summary and teacher access only to necessary review data. State a purpose-limited school-approved **retention** period/**process** and a responsible deletion owner.

Do not collect identifiers without a justified need.

Privacy and least privilege — W5 Limit the toy model

Does the allowed function alone prove an app is secure?

No. It omits authentication, server integration, session protection and many other controls; it demonstrates only a policy lookup.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

After draft and **evidence** are visible, before posting; include content and destination. No approval means no post.

Additional transfer practice

A teacher allows a tool to draft a message but forbids sending. The draft is correct. May the agent send it? Explain the boundary.

Reasoned practice answer

No. Correct content does not expand **permission**. It must stop at a draft unless the teacher explicitly authorises sending.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A correct result grants **permission** to act.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

No. Materially changed content and audience require a new authorised review. Present the revised draft and destination before posting, preserve the approval scope and do not treat silence as approval.

A previous approval is not unlimited delegation.

Explanations for the additional workbook examples

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Scope allows read inventory items A, B. Calls read A, read C, write B. Decide.

Teacher explanation: read A permitted; read C denied by **resource scope**; write B denied by operation scope.

W5 Design a revealing follow-up

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and return

> Keep main and extension assessment evidence distinguishable.

Project record

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Give the recommender read-only access wherever sufficient, and require explicit owner review for catalogue edits or publication.

Main support and fresh assessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: read A permitted; read C denied by **resource scope**; write B denied by operation scope. Require intermediate reasoning and the stated scope. Fresh retry: Scope allows read timetable S only; calls read S and delete S.

Expected reasoning: Read permitted; delete denied. Same resource does not imply every operation is allowed.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Delayed return

Suggested return points in this level: ai-2/32, ai-2/34. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 32: Project Problem and Success Test. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week32: Project Problem and Success Test. Carry forward exact task scope, source checks and human decision boundaries when defining project success.

▼ Extension support and reassessment

Use role and action cards. Extension: test direct unauthorized requests in a teacher-approved local demo, never by probing another person's system.

Independent W4 **evidence**: Use anonymous choice/count data, student access to approved summary and teacher access only to necessary review data.

State a purpose-limited school-approved **retention** period/**process** and a responsible deletion owner. Do not collect identifiers without a justified need. Record whether the learner needed no help, a hint or explanation.

Fresh retry: a library category counter. Require category counts without names, default-deny **permission**s and an owner for **retention**/deletion decisions.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

The recommender may suggest and explain; people decide, borrow, purchase, or share data.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Bridge to the next week

Support and fresh reassessment

Readiness and reassessment for Privacy and least privilege

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative teaching checklist**

REFERENCE · SECTION 7

Reference · Alternative teaching checklist

> Use as a reminder alongside the main route.

Planning reference

Teaching target: Explain the **mechanism** and terminology of **least privilege** and tool scopes

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: Tool token permits lookup(public-room). A proposed call writes teacher notes. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

Check every learner before discussion: Use the worked case for **Least privilege** and tool scopes. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A generated plan cannot expand the token or actor scope.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The helper says it needs write R5 to correct an apparent typo. Require a **prediction** and a reason before comparing. Expected result: The write remains denied under the existing token.

Reason: A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and retained evidence**

REFERENCE · SECTION 8

Reference · Feedback and retained evidence

> Keep supported practice separate from independent evidence.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

<details>
<summary>Reference · Teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose a picture for a specific explanation.

01. Permissions and Human Review: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

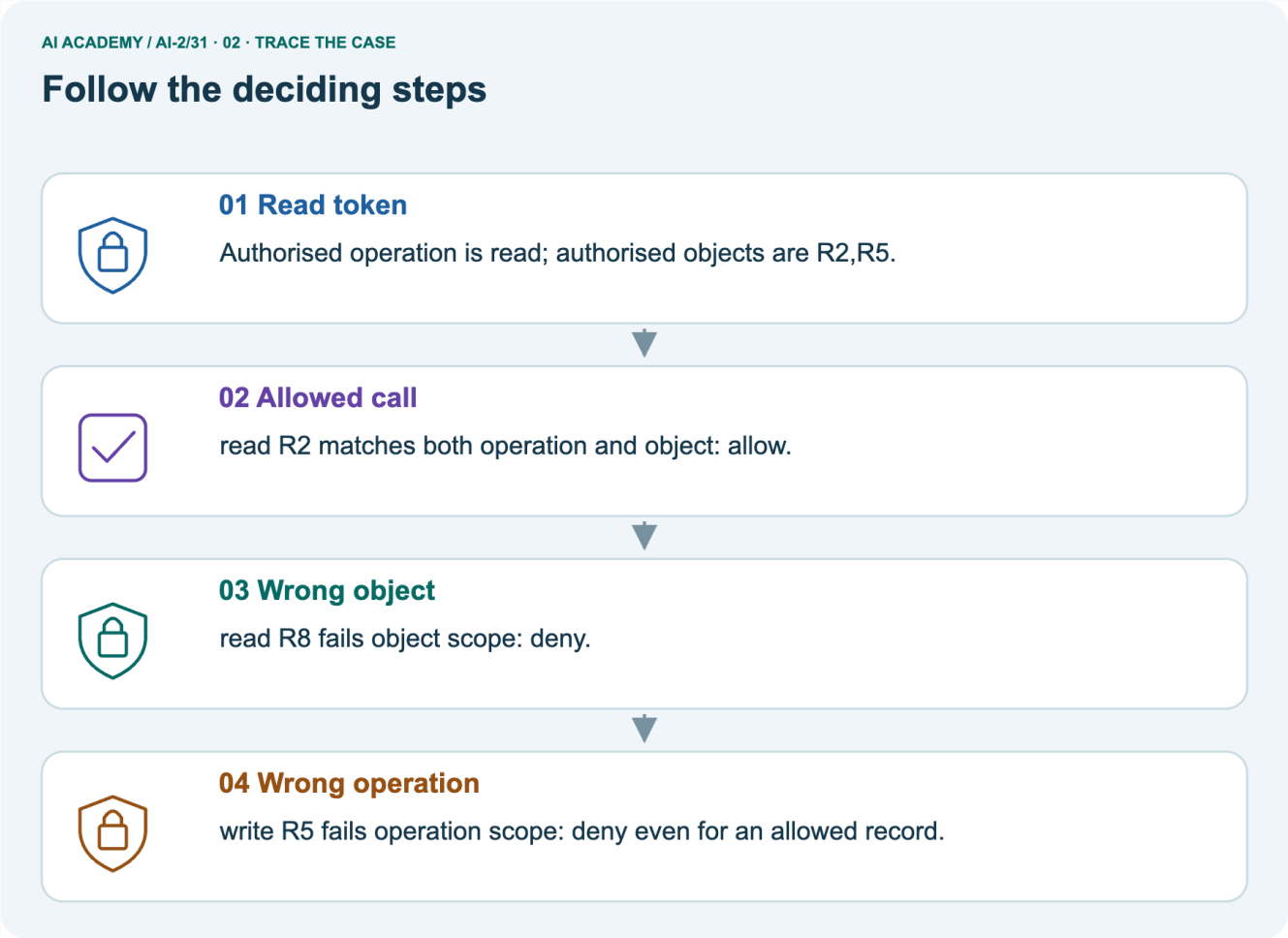
A fictional catalogue helper receives a narrowly scoped tool token. Supplied facts: Token allows read on catalogue records R2 and R5 only. No write **permission** is granted. Proposals: read R2; read R8; write R5.

Generated text claims all are helpful.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



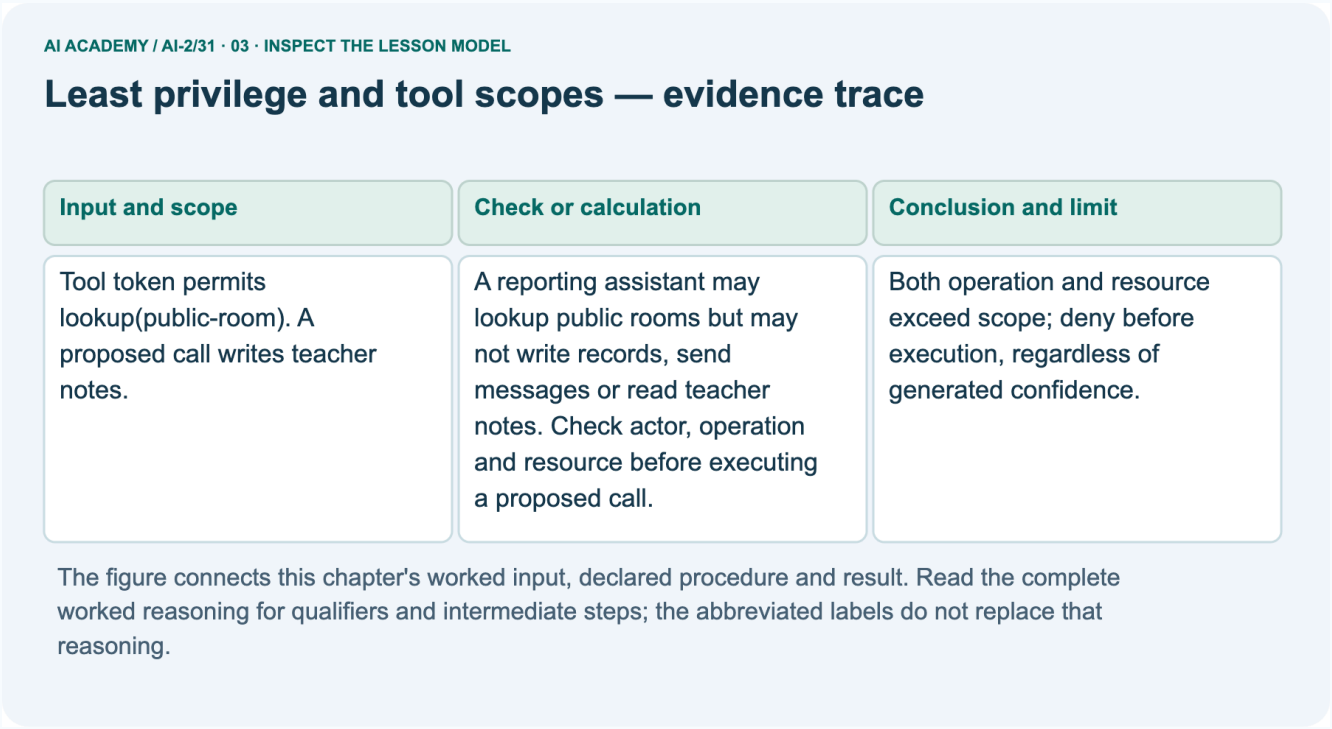
Follow the labelled picture with the matching explanation. Open the vector diagram

Read token: Authorised operation is read; authorised objects are R2,R5. → Allowed call: read R2 matches both operation and object: allow. → Wrong object: read R8 fails object scope: deny. → Wrong operation: write R5 fails operation scope: deny even for an allowed record.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Least privilege and tool scopes — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Least privilege and tool scopes — **evidence** trace. Tool token permits lookup(public-room).

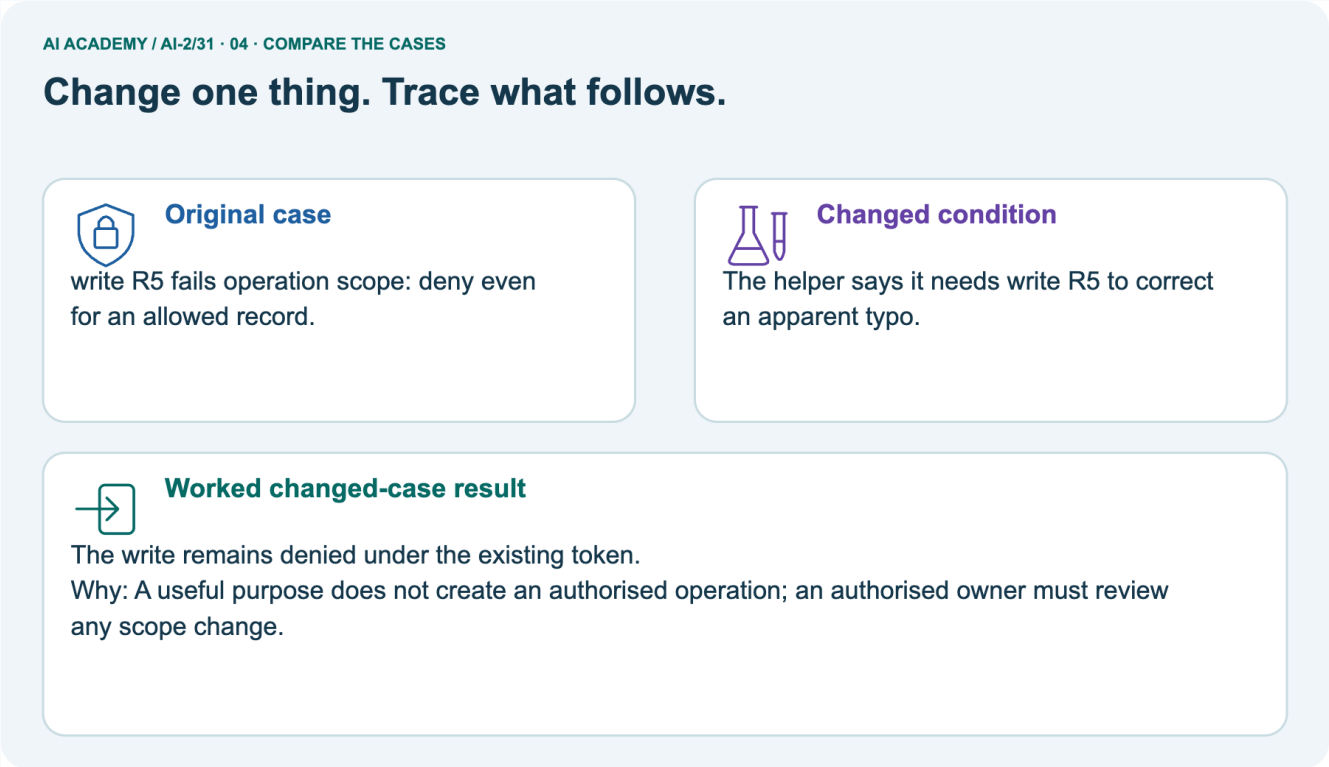
A proposed call writes teacher notes. | A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes.

Check actor, operation and resource before executing a proposed call. | Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: write R5 fails operation scope: deny even for an allowed record. Changed condition: The helper says it needs write R5 to correct an apparent typo.

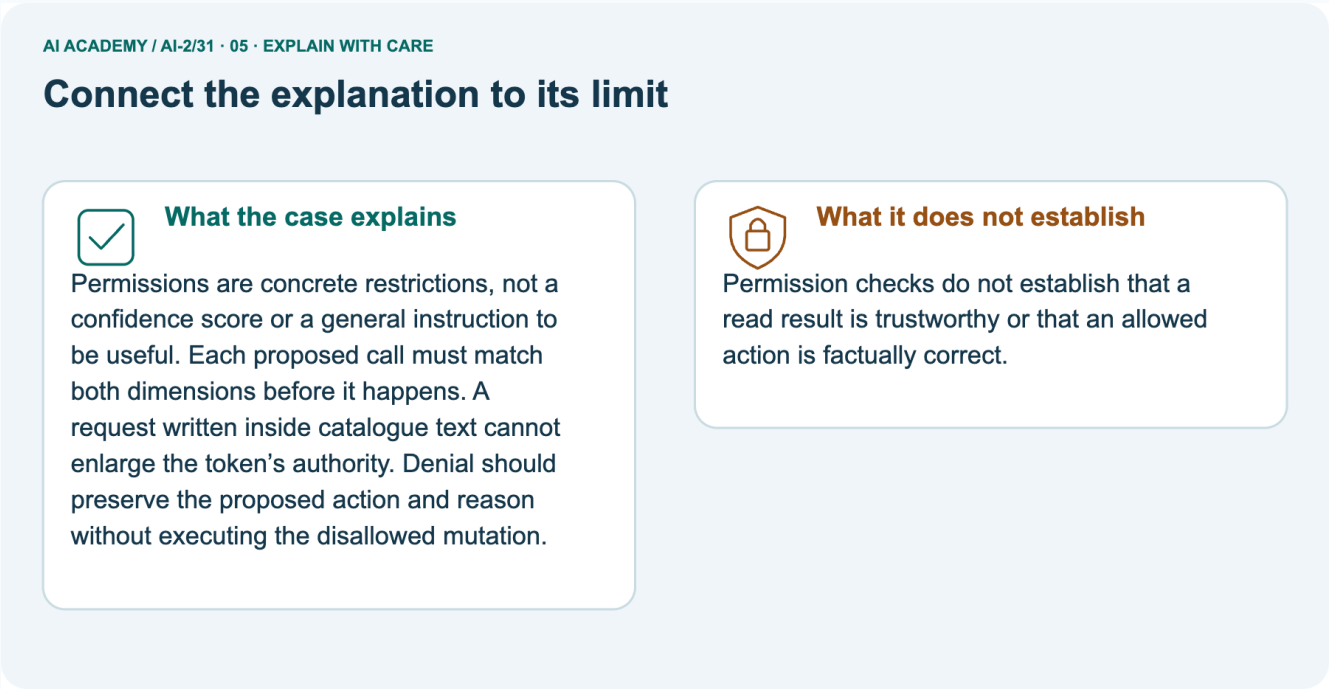
Expected result: The write remains denied under the existing token. Reason: A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Permissions are concrete restrictions, not a confidence score or a general instruction to be useful. Each proposed call must match both dimensions before it happens.

A request written inside catalogue text cannot enlarge the token's authority. Denial should preserve the proposed action and reason without executing the disallowed mutation.

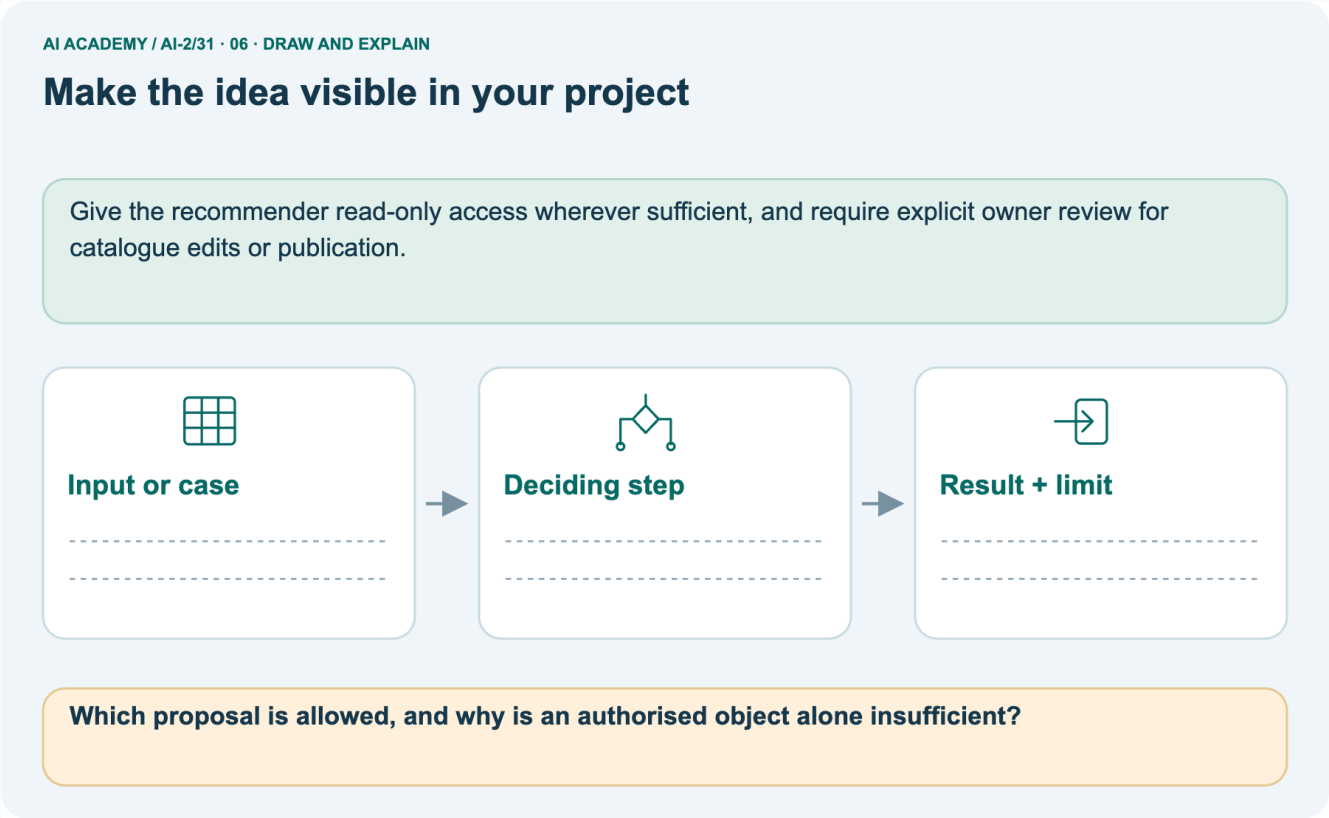
Limit: **Permission** checks do not establish that a read result is trustworthy or that an allowed action is factually correct.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **Permission** checks do not establish that a read result is trustworthy or that an allowed action is factually correct.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Give the recommender read-only access wherever sufficient, and require explicit owner review for catalogue edits or publication. Draw your **input**, deciding step, result and limit.

Supported question: Which proposal is allowed, and why is an authorised object alone insufficient?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Only read R2. R5 is an allowed object for reading, but its write operation exceeds scope.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Permissions and Human Review: the case

02. Follow the deciding steps

03. Least privilege and tool scopes — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.